

Overview (step 1):

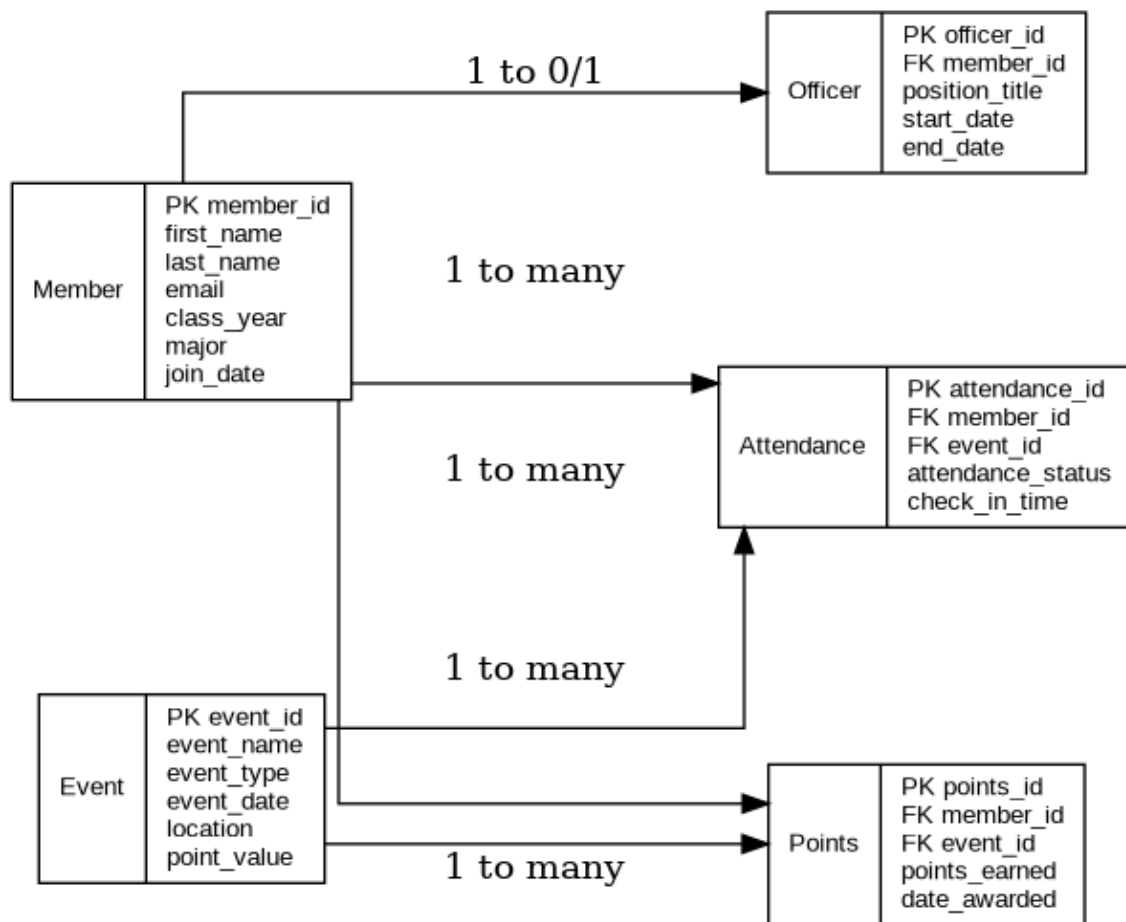
The Phi Mu Event & Points Tracker is a database for a sorority chapter that can help track member involvement, event attendance and point totals. Officers want to easily track who went to what events and how many points each member has accrued over the semester. This would replace an unorganized Texts and spreadsheets system. Officers will be able to interact with the database to enter members, enter events, input attendance, and update points. Members can interact with the database to see what they should be doing to meet chapter expectations by seeing their attendance. The intended audiences who will use this database are chapter officers, chapter members, chapter advisor, and technology/admin chair. Chapter officers will need to enter new event information, track who attended said events, and input how many points attendees receive. Chapter members will need to see what events are occurring and what their point total is. Chapter advisors will possibly need reports to see how members are participating. The technology/admin chair will need to maintain this database and ensure information is up to date. MySQL Workbench will be used as the RDBMS. Everything will be built on a laptop/desktop computer. Since this is a class project, we will only be able to display the front-end application through mockup screens. These screens will be demonstrated through screenshots of the forms used to add members, events, input attendance, and view points reports. The data included in this database will be member names, member emails, member class years, member officer positions, event names, event dates, attendance, and points. Sample make-up data will be used as starter data. In an actual chapter, this data would be entered manually after each event or week. Several procedures will guide the system. Officers will add new members at the beginning of the semester, create events before they happen, record attendance after each event, and update points based on participation. Reports can be created throughout the semester to show which members are active and which members may need reminders. This project assumes that the system is being tested by one sorority chapter first. It also assumes that only approved officers can edit the database, while general members can only view limited information about their own attendance and points. The project will use a pilot deployment because the system should first be tested with a small group of officers before being used by the entire chapter. The main entities for this database are Member, Officer, Event, Attendance, and Points. These entities allow the system to track who the members are, which members hold

leadership roles, what events are created, who attends each event, and how many points each member earns. This supports the project requirements because the database includes people, procedures, data sources, assumptions, and a selected RDBMS.

List of external tools & Sources

- I used Mysql Workbench to create the database
- I used Canva to my a mock up of my website
- I used Chatgpt to give me data
- I used ERDplus to create my ERDs and DataBase Modeling

ERD:



Database Modeling (step 2):

Member	Officer	Event	Attendance	Points
<u>Member_id</u>	<u>Officer_id</u>	<u>Event_id</u>	<u>Attendance_id</u>	<u>Points_id</u>
First_Name	Member_id	Event_Name	Member_id	Member_id
Last_Name	Position_Title	Event_Type	Event_id	Event_id
Email (U)	Start_Date	Event_Date	Attendance_Status	Points_Earned
Class_Year	End_Date	Event_Location	Check_In_Time	Date_Awarded
Major		Point_Value		
Join_Date				

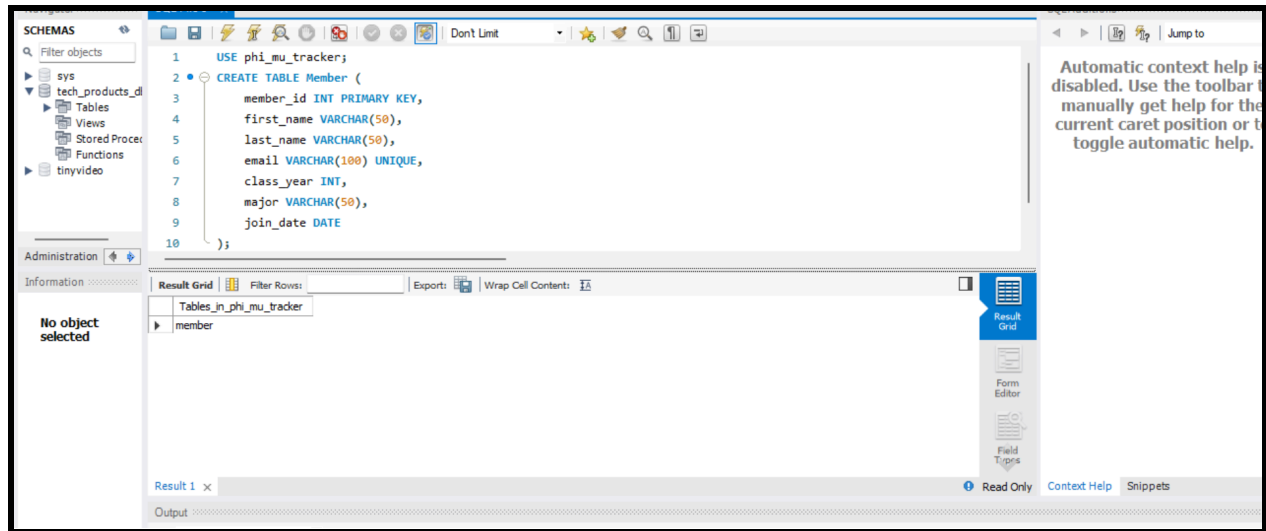
Database:

Created Table:

The screenshot displays a database management tool interface. On the left, a 'SCHEMAS' pane shows a tree view with 'sys' and 'tech_products' (containing 'Tables', 'Views', 'Stored Proc', and 'Functions'). Below this is an 'Administration' section with a 'No object selected' message. The main workspace shows a SQL editor with the command 'USE phi_mu_tracker;'. To the right of the editor is a 'Context Help' pane with a message: 'Automatic context help disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.' Below the editor is an 'Output' pane with a table showing the execution results:

#	Time	Action	Message	Duration / Fetch
2	19:35:48	USE phi_mu_tracker	0 row(s) affected	0.000 sec
3	19:35:48	USE phi mu tracker	0 row(s) affected	0.000 sec

At the bottom left, there are tabs for 'Object Info' and 'Session'.



Basic Structure of Table

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- sys
- tech_products_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
- tinyvideo

Administration

Information

No object selected

SQL File 3*

```
41 points_id INT PRIMARY KEY,
42 member_id INT,
43 event_id INT,
44 points_earned INT,
45 date_awarded DATE,
46 FOREIGN KEY (member_id) REFERENCES Member(member_id),
47 FOREIGN KEY (event_id) REFERENCES Event(event_id)
48 );
49 SHOW TABLES;
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Filter Rows

Export

Wrap Cell Contents

Tables_in_phi_mu_tracker

- attendance
- event
- member
- officer
- points

Result 5

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	19:43:16	CREATE TABLE Member (member_id INT PRIMARY KEY, first_name VARCHAR(50)...	Error Code: 1050. Table 'member' already exists	0.000 sec
22	19:43:19	SHOW TABLES	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Table with Data: Members, Attendance, Events, Officer, Points

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

member

Columns

Indexes

Foreign Keys

Triggers

officer

Columns

Indexes

Foreign Keys

Triggers

points

Columns

Indexes

Foreign Keys

Triggers

Views

Stored Procedures

Functions

sys

tech_products_db

tinyvideo

Administration Schemas

Information

HOST: localhost

Port: 3306

Login User: root

SQL File 3* member

1 • SELECT * FROM phi_mu_tracker.member;

Result Grid

member_id	first_name	last_name	email	class_year	major	join_date
1	Ava	Johnson	ava.johnson@cnu.edu	2026	Marketing	2024-09-01
2	Mia	Smith	mia.smith@cnu.edu	2027	Psychology	2024-09-01
3	Emma	Brown	emma.brown@cnu.edu	2025	Business	2023-09-01
4	Olivia	Davis	olivia.davis@cnu.edu	2026	Communication	2024-01-15
5	Sophia	Wilson	sophia.wilson@cnu.edu	2027	Information Science	2023-09-01
6	Grace	Miller	grace.miller@cnu.edu	2025	Biology	2023-09-01
7	Ella	Taylor	ella.taylor@cnu.edu	2026	English	2024-01-15
8	Lily	Moore	lily.moore@cnu.edu	2027	Sociology	2024-09-01
*	NULL	NULL	NULL	NULL	NULL	NULL

member 1 x

Apply Revert

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

phi_mu_tracker

Tables

attendance

Columns

Indexes

Foreign Keys

Triggers

event

Columns

Indexes

Foreign Keys

Triggers

member

Columns

Indexes

Foreign Keys

Triggers

officer

Columns

Indexes

Foreign Keys

Triggers

Administration Schemas

Information

HOST: localhost

Port: 3306

Login User: root

Current User: root@localhost

SSL cipher: SSL not used

Server

Product: MySQL Commu

SQL File 3* member attendance

1 • SELECT * FROM phi_mu_tracker.attendance;

Result Grid

attendance_id	member_id	event_id	attendance_status	check_in_time
1	1	1	Present	18:00:00
2	2	1	Present	18:05:00
3	3	1	Absent	NULL
4	4	2	Present	19:00:00
5	5	2	Late	19:10:00
6	6	2	Present	19:02:00
7	7	3	Present	17:00:00
8	8	3	Absent	NULL
9	1	3	Present	17:05:00
10	2	2	Present	19:03:00
*	NULL	NULL	NULL	NULL

attendance 1 x

Apply Revert

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

phi_mu_tracker

- Tables
 - attendance
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
 - event
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
 - member
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
 - officer
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers

Administration Schemas

Information

HOST: localhost

Port: 3306

SQL File 3* member attendance event x

1 • SELECT * FROM phi_mu_tracker.event;

Result Grid

event_id	event_name	event_type	event_date	event_location	point_value
1	Bid Day Celebration	Social	2025-09-01	Phi Mu House	10
2	Philanthropy Night	Philanthropy	2025-09-10	Student Center	15
3	Chapter Meeting	Meeting	2025-09-05	Lecture Hall	5
* NULL	NULL	NULL	NULL	NULL	NULL

Form Editor

Field Types

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

phi_mu_tracker

- Tables
 - attendance
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
 - event
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
 - member
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
 - officer
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers

Administration Schemas

Information

HOST: localhost
Port: 3306
Login User: root
Current User: root@localhost

SQL File 3* member attendance event officer x

1 • SELECT * FROM phi_mu_tracker.officer;

Result Grid

	officer_id	member_id	position_title	start_date	end_date
1	1	1	President	2025-01-01	2025-12-01
2	3	3	Vice President	2025-01-01	2025-12-01
3	5	5	Treasurer	2025-01-01	2025-12-01
*	NULL	NULL	NULL	NULL	NULL

Form Editor

Field Types

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

member

- Columns
- Indexes
- Foreign Keys
- Triggers

officer

- Columns
- Indexes
- Foreign Keys
- Triggers

points

- Columns
- Indexes
- Foreign Keys
- Triggers

Views

Stored Procedures

Functions

sys

tech_products_db

tinyvideo

Administration Schemas

Information

HOST: localhost
Port: 3306
Login User: root
Current User: root@localhost
SSL cipher: SSL not used

SQL File 3* member attendance event officer points x

1 • SELECT * FROM phi_mu_tracker.points;

Result Grid

	points_id	member_id	event_id	points_earned	date_awarded
1	1	1	1	10	2025-09-01
2	2	1	1	10	2025-09-01
3	4	2	2	15	2025-09-10
4	5	2	10	10	2025-09-10
5	6	2	15	15	2025-09-10
6	7	3	5	5	2025-09-05
7	1	3	5	5	2025-09-05
8	2	2	15	15	2025-09-10
9	3	1	0	0	2025-09-01
10	8	3	0	0	2025-09-05
*	NULL	NULL	NULL	NULL	NULL

points 1 x

Apply Revert

Queries:

Event Information & Points its Worth

The screenshot shows the MySQL Workbench interface with a query editor window titled "Phi_Mu_Tracker". The query is as follows:

```
1 • SELECT * FROM phi_mu_tracker.points;
2 • SHOW TABLES;
3 • SELECT * FROM Member;
4 • SELECT * FROM Event;
```

The "Result Grid" displays the following data:

event_id	event_name	event_type	event_date	event_location	point_value
1	Bid Day Celebration	Social	2025-09-01	Phi Mu House	10
2	Philanthropy Night	Philanthropy	2025-09-10	Student Center	15
3	Chapter Meeting	Meeting	2025-09-05	Lecture Hall	5
NULL	NULL	NULL	NULL	NULL	NULL

Name & Position

The screenshot shows the MySQL Workbench interface with a query editor window titled "Phi_Mu_Tracker". The query is as follows:

```
7
8
9 • SELECT
10     Member.first_name,
11     Member.last_name,
12     Officer.position_title
13 FROM Officer
14 JOIN Member ON Officer.member_id = Member.member_id;
```

The "Result Grid" displays the following data:

first_name	last_name	position_title
Ava	Johnson	President
Emma	Brown	Vice President
Sophia	Wilson	Treasurer

Names of People who were Absent/Present at Event (Title Listed)

The screenshot shows the MySQL Workbench interface with a query window titled 'Phi_Mu_Tracker'. The query is as follows:

```
17 SELECT
18     Event.event_name,
19     Member.first_name,
20     Member.last_name,
21     Attendance.attendance_status
22 FROM Attendance
23 JOIN Member ON Attendance.member_id = Member.member_id
24 JOIN Event ON Attendance.event_id = Event.event_id;
```

The Result Grid displays the following data:

event_name	first_name	last_name	attendance_status
Bid Day Celebration	Ava	Johnson	Present
Bid Day Celebration	Mia	Smith	Present
Bid Day Celebration	Emma	Brown	Absent
Philanthropy Night	Olivia	Davis	Present
Philanthropy Night	Sophia	Wilson	Late
Philanthropy Night	Grace	Miller	Present
Philanthropy Night	Mia	Smith	Present
Chapter Meeting	Ella	Taylor	Present
Chapter Meeting	Lily	Moore	Absent
Chapter Meeting	Ava	Johnson	Present

People With The Highest Points

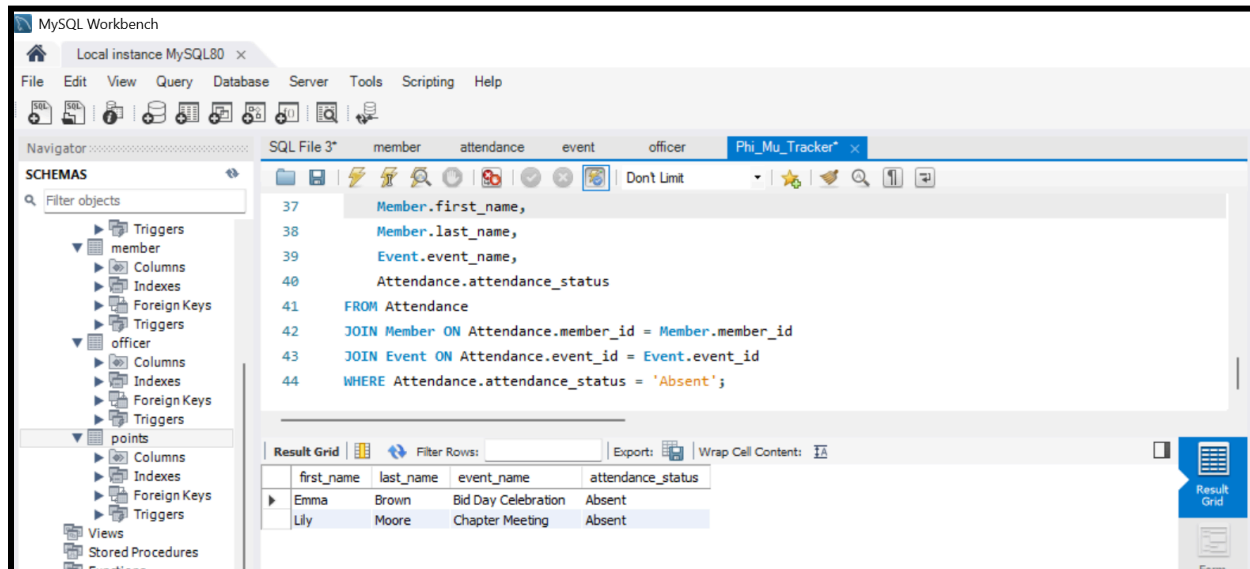
The screenshot shows the MySQL Workbench interface with a query window titled 'Phi_Mu_Tracker'. The query is as follows:

```
27 SELECT
28     Member.first_name,
29     Member.last_name,
30     SUM(Points.points_earned) AS total_points
31 FROM Points
32 JOIN Member ON Points.member_id = Member.member_id
33 GROUP BY Member.member_id, Member.first_name, Member.last_name
34 ORDER BY total_points DESC;
```

The Result Grid displays the following data:

first_name	last_name	total_points
Mia	Smith	25
Ava	Johnson	15
Olivia	Davis	15
Grace	Miller	15
Sophia	Wilson	10
Ella	Taylor	5
Emma	Brown	0
Lily	Moore	0

Who was ONLY Absent



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with a search filter. The main editor window contains a SQL query that joins the 'member', 'attendance', and 'event' tables to find members who are absent. The query is as follows:

```
37 Member.first_name,  
38 Member.last_name,  
39 Event.event_name,  
40 Attendance.attendance_status  
41 FROM Attendance  
42 JOIN Member ON Attendance.member_id = Member.member_id  
43 JOIN Event ON Attendance.event_id = Event.event_id  
44 WHERE Attendance.attendance_status = 'Absent';
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query. The grid has four columns: first_name, last_name, event_name, and attendance_status. The results are as follows:

first_name	last_name	event_name	attendance_status
Emma	Brown	Bid Day Celebration	Absent
Lily	Moore	Chapter Meeting	Absent

Front End Application (Step 4):

Phi Mu Tracker

Add New Member

Member Information

First Name: _____

Last Name: _____

Email: _____

Academic Information

Class Year: _____

Major: _____

Join Date: _____

[Submit]

Phi Mu Tracker

Create Event

Event Name: _____

Event Type: _____

Event Date: _____

Location: _____

Point Value: _____

[Create Event]

Phi Mu Tracker

Record Attendance

Select Event: _____

Member Name: _____

Attendance Status:

Present Absent Late

[Save Attendance]

Phi Mu Tracker

Member Points Leaderboard

Sorted by highest total points

Name	Total Points
Ava Johnson	15
Mia Smith	25
Emma Brown	10
Olivia Davis	15
Sophia Wilson	10

Database Deployment, Use, & Maintenance (Step 5, 6, 7):

DATABASE DEPLOYMENT

The Phi Mu Event & Points Tracker will use a pilot deployment approach. This means the system will first be tested by a small group of chapter officers before being introduced to the entire chapter. The purpose of using a pilot deployment is to identify any issues, errors, or usability concerns in a controlled environment before full implementation. By starting with a smaller group, the system can be improved based on feedback, making it more reliable and effective when it is eventually used by all members. This approach reduces risk and ensures the database functions correctly before full adoption.

DATABASE USE (USER INSTRUCTIONS)

The Phi Mu Event & Points Tracker is designed to be simple and easy for users to operate. Chapter officers will begin by adding new members into the system using the “Add New Member” form. Next, officers can create events using the “Create Event” screen by entering event details such as name, date, and point value. After each event, officers will use the “Record Attendance” screen to mark which members were present, absent, or late.

Once attendance is recorded, the system automatically allows tracking of participation and points. Officers and members can view the “Member Points Leaderboard” to see total points earned and compare participation levels. This system helps ensure accountability and encourages member involvement in chapter activities.

DATABASE ADMINISTRATION & MAINTENANCE

The database will be maintained by designated chapter officers, such as the technology chair or president. Only authorized users will have permission to add, update, or delete data in the system to ensure accuracy and security. General members will have limited access, typically restricted to viewing their own participation and point totals.

To maintain data integrity, the database should be backed up regularly, such as on a weekly basis. This prevents data loss in case of technical issues. Additionally, access permissions should be updated each semester to remove former officers and grant access to new leadership. Member information, such as email addresses, must be kept confidential and should not be shared outside of the organization.

All tools used in this project include MySQL Workbench for database development, ERDPlus for diagram creation, and Canva for front-end mockups.